

Note-Vorticity Boundary Condition and Related Issues for Finite Difference Schemes

Information

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Summary of the content

In this article, the authors presented the $\omega - \psi$ formulation of the Navier-Stokes Equation and numerical schemes to determine the vorticity boundary condition.

The local formulation of ω_{BC} was given by Thom in 1933, which can be stated as, for example,

$$\omega_{0,j} = \frac{2}{\Delta x^2} \psi_{1,j}$$

There are two global formulations of ω_{BC} : *Quartapelle's Vorticity Boundary Conditions* and *Anderson's Vorticity Boundary Condition*.

- Quartapelle's method uses the orthogonality of ω to the space of harmonic functions (denoted by $\mathcal{H}$) on the domain Ω , with respect to the standard L^2 inner product.
- Anderson's method uses the property of inverse operator of Laplacian to deduce the boundary condition with the interior vorticity field, such that

$$\begin{aligned} D_n \Delta_0^{-1} (-(\mathbf{u} \cdot \nabla) \omega + \nu \Delta \omega) &= 0 \\ \Rightarrow D_n \Delta_0^{-1} (-(\mathbf{u} \cdot \nabla) \omega_{bd} + \nu \Delta \omega_{bd}) &= -D_n \Delta_0^{-1} (-(\mathbf{u} \cdot \nabla) \omega_{int} + \nu \Delta \omega_{int}) \end{aligned}$$

where

$$D_n(\cdot) = \frac{\partial(\cdot)}{\partial n} = \nabla(\cdot) \cdot \mathbf{n}$$

and the subscript 0 means homogeneous Dirichlet boundary condition is taken. (Thus, the boundary conditions of ψ are satisfied in Anderson's method.)

What do I learn/understand ?

- Anderson's method starts solving the problem with interior mesh points, and the boundary condition is merely considered as an auxiliary variable to obtain the vorticity in the interior mesh points as time varying. However, the usage of inverse Laplacian operator is quite interesting, because it is an application of its non local property, which can be used to establish a relation of the distribution of ω in the domain and its the boundary condition.
- Quartapelle's method is different from Anderson's method, since it does use the boundary condition ω_{BC} to solve the vorticity field of the interior mesh points. But it is also complicate to apply because it is based on the orthogonality of ω . Thus, it requires knowing all discrete harmonic functions and/or discrete Green's function at all boundary points.
- Anderson's method can be converted to a local form (*Fromm's Formula*) by some particular discrete expressions. However, we cannot conclude that local form can replace the global vorticity condition, because in some cases the global vorticity conservation is not guaranteed, and it might fail (for example, Hung and Kinney, 1988).

Related Articles

- L. Quartapelle and F. Valz-Gris, Projection conditions on the vorticity in viscous incompressible flows, Int. J. Numer. Meth. Fluids 1 (1981) 129–144.
- C.R. Anderson, Vorticity boundary conditions and boundary vorticity generation for two-dimensional viscous incompressible flows, J. Comput. Phys. 80 (1989), 72–97.
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- Hung, S. C., and Kinney, R. B. (1988). Unsteady viscous flow over a grooved wall: A comparison of two numerical methods. Int. J . Num. Meth. FI. 8, 1403.